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Pluvio Grow — M5 Build

RAIN, WITH A BRAIN.



A 13-plant, weighed-dose plant waterer from M5Stack parts

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Build a 13-Plant Self-Watering Robot (Open Source) — Pluvio Grow M5

Intro

I kept killing my plants — so I built a little robot that waters 13 of them, each its own exact amount, measured **by weight**. One arm spins around a carousel, stops at each plant, and doses it from a single reservoir. It runs fully offline on a tiny ESP32, and I open-sourced the whole thing so you can build your own.

What makes it different from a $40 drip timer: it *weighs* every pour with a load cell, so the cactus gets 15 g and the fern gets 60 g — not "everything for 30 seconds." Rain, with a brain.

- **Difficulty:** intermediate (soldering/crimping, a stepper tweak, 3D printing)
- **Time:** a weekend
- **Cost:** roughly **$80–120** in parts (see notes on min-order quantities)

Supplies

Electronics (DigiKey)

Part	MPN	Qty	Unit
M5Stack **AtomS3R** dev kit (ESP32-S3 + 0.85″ display)	C126	1	$17.50
M5Stack **Mini Scales Unit** (HX711 load cell)	U177	1	$6.95
M5Stack **Atomic Motion Base** v1.2 (stepper + servo driver)	A090-V12	1	$8.50
Olimex SM-5VDC stepper (5-wire; convert to 4-wire **bipolar** — see note)	SM-5VDC	1	$1.76
AS5600 magnetic rotary encoder (arm position)	AliExpress	1	—
5 V 5 W wall adapter	PSA05A-050QL6-H	1	$3.91
USB-A → USB-C cable	DH-20M50053	1	$1.92
Connector housing, 2-pos 2.54 mm	M20-1060200	2	$0.10
Crimp socket, 22–30 AWG gold	M20-1180042	2	$0.09

> **Heads-up on DigiKey min-order:** several of these have minimum quantities (e.g. the stepper and adapter ship in packs), so your cart total will exceed the per-unit cost above. Buy one device's worth; you'll have spares.

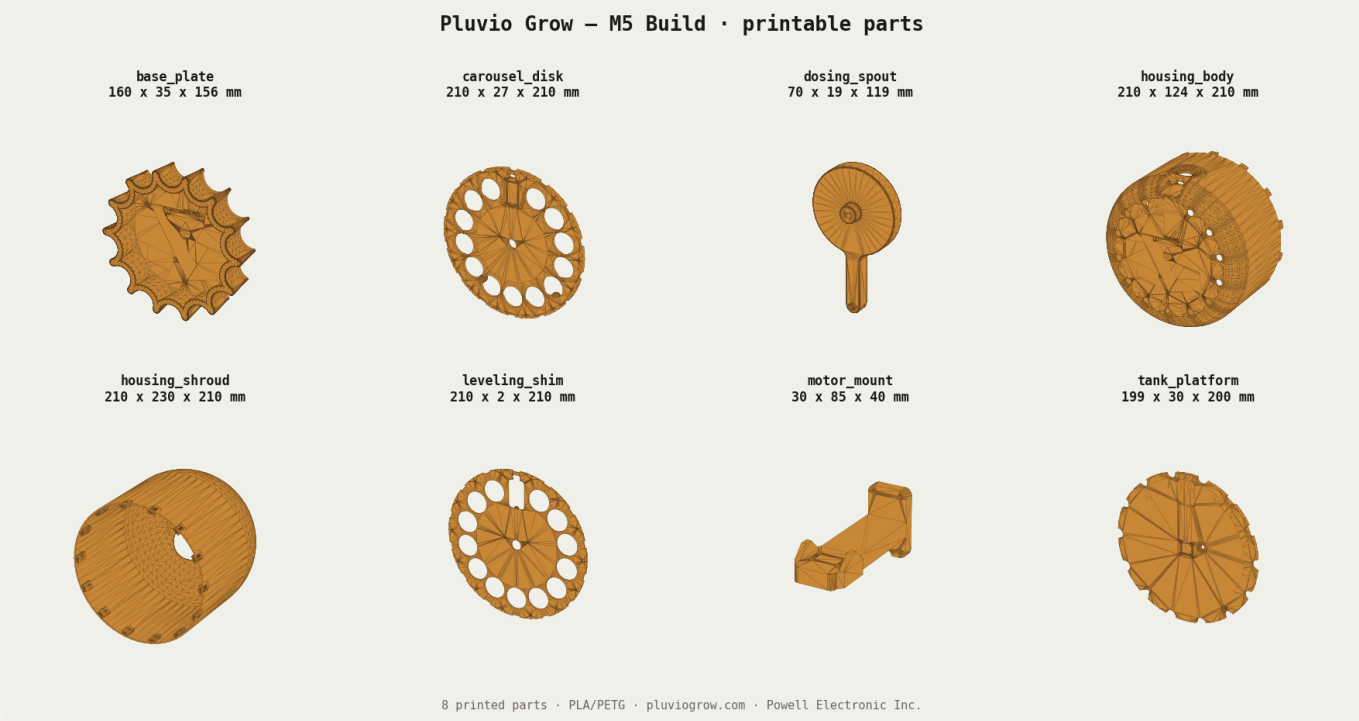
Mechanical & plumbing

Part	Source	Notes
Solenoid valve, 5 V	AliExpress	the dosing valve
Solenoid driver (RC-PWM → DC)	AliExpress	lets the Atomic Motion servo channel switch the valve
60 mL Luer-lock syringe ×13	generic (AliExpress / Amazon)	per-plant carousel vessels
Luer-lock → 1/4″ tube adapter ×13	generic	connects the tube to each syringe
Silicone tubing, 1/8″ ID × 1/4″ OD, soft 50A	McMaster 51135K16	one run per plant — ⚠️ **CONFIRM total length for your layout**
1-gallon utility jug	Uline S-10746B	reservoir
Top-hat rubber grommets (6 mm hole / 4 mm ID)	Amazon (uxcell 46-pc)	valve / tube seal
M3 × 8 mm thread-forming screws	McMaster 99461A941	plastic assembly
M3 × 20 mm thread-forming screws	McMaster 99461A947	plastic assembly

> ⚠️ **CONFIRM — not yet in the BOM:** a **diametric magnet** for the AS5600 (glued to the arm's rotation axis — the encoder reads it), and the **tube length** for your layout.

3D-printed parts

Eight printed parts, included in `hardware/stl/`, from the confirmed M5 assembly (`hardware/step/pluvio_m5_assembly.STEP`). Print in PLA/PETG, 0.2 mm layers, 3–4 walls, 20–30% infill.



Tools

Soldering iron, wire crimpers (for the Harwin sockets), 3D printer, small screwdriver, hobby knife, a cheap digital scale (to sanity-check the load cell).

Step 1 — Print the parts

Print all eight parts from `hardware/stl/`: `carousel_disk` (13 metering positions), `housing_body`, `housing_shroud`, `base_plate`, `tank_platform`, `dosing_spout`, `motor_mount`, and `leveling_shim`. PLA or PETG, 0.2 mm layers, 3–4 walls, 20–30% infill. Most parts need no supports; orient the shroud and housing open-side-down. See `hardware/stl/README.md` for the full size table.

Step 2 — Wire the electronics

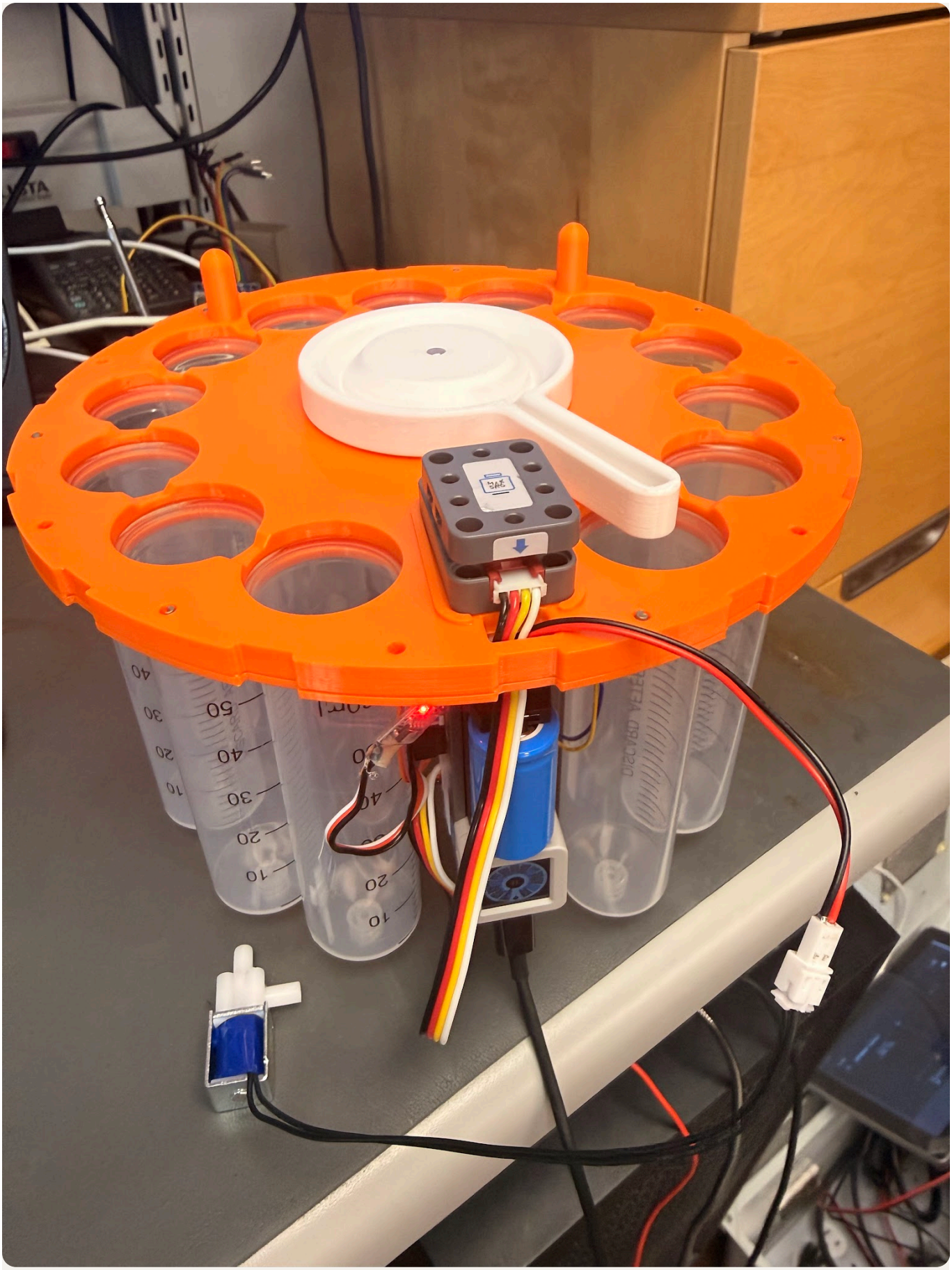
The **AtomS3R** sits on the **Atomic Motion Base**, which drives both the motor and the valve, so there's no separate motor driver board.

- **Encoder → Port B** and **Mini Scales → Port C** — these are **Grove** connectors: just plug the cable in, no soldering.
- **Stepper → Atomic Motion M1/M2** — convert to bipolar first (note below), then coil A → M1, coil B → M2 (crimp the Harwin sockets into the 2-pos housings).
- **Solenoid valve → driver module → "Servo 1"** — the firmware opens/closes the valve by writing servo angles to the driver.
- **Power** via the 5 V adapter / USB-C.

Bipolar mod (important): the Olimex SM-5VDC ships as a stock **5-wire unipolar** 28BYJ-48. To drive it from the M5 H-bridge, convert it to a **4-wire bipolar** motor: open the motor and disconnect the shared common wire from the two internal coil pairs, so each coil is independent. Then coil A → M1, coil B → M2.

Step 3 — Assemble the mechanism

Mount the motor and carousel, fit the encoder magnet on the rotation axis, seat the per-plant vessels, and plumb the dosing spout. Seal the valve/tube passes with the top-hat grommets.



Step 4 — Plumb it

Run one tube from the carousel spout to each plant; feed the pump/valve from the 1-gallon jug. Keep the reservoir at or below the unit so a closed valve can't siphon.

Step 5 — Flash the firmware

Open-source, offline, no account. Full steps in `firmware/README.md`: install the ESP32 core + four libraries in the Arduino IDE, set the board, upload. On first boot it opens a `Pluvio-Setup-…` Wi-Fi hotspot — join it, pick your network, and manage the unit from its own web page.

Step 6 — Calibrate & first water

Run the carousel calibration (it drives into both endstops to learn zero), set each plant's dose and interval, fill the jug, and let it run.

Credits & license

Open-sourced by **Pluvio Grow**, a product of **Powell Electronic Inc.** Hardware & docs under **CC BY-NC-SA 4.0**; patent pending (US 64/071,683) on the weighed-dosing mechanism — personal/non-commercial builds welcome. Build one, show us, tag us. 🌱